Experiment 6: One-Pass Stream Processing

Aim

To process a data stream in a single pass by reading every value exactly once, updating the count, running sum and running average immediately as each value arrives, and finally displaying the total elements processed, the final sum, the final average and the execution time.

Theory

One-pass stream processing means that every element of the stream is read only once and is processed immediately on arrival. The complete data is never stored or scanned a second time, so only a few summary variables such as count and total need to be kept in memory. From these two variables the running average can be derived at any moment. This makes the method memory efficient and suitable for continuous or very large data where storing the entire stream is not practical. The time module is used to record the start and end time so that the execution time of the whole stream can be measured.

Requirements

- Python 3.x
- Jupyter Notebook
- Time module

Step 1: Import Library and Define Stream Data

```
import time

stream_data = [5, 9, 11, 12, 15]
```

Interpretation: Imports the time module for measuring execution time and defines the incoming stream of values.

Step 2: Process Stream Data

```
count = 0
total = 0
start_time = time.time()

for value in stream_data:
    count += 1
    total += value
    running_average = total / count
    print("Received:", value)
    print("Running sum:", total)
    print("Running avg:", running_average)

end_time = time.time()
```

Interpretation: Reads each value once. The count and total are updated immediately and the running average is recalculated after every arrival. The start and end times are recorded around the loop.

Step 3: Display Final Results

```
print("Total elements processed:", count)
```

```
print("Final sum:", total)
print("Final average:", total / count)
print("Execution time:", round(end_time - start_time, 6), "seconds")
```

Interpretation: Displays the final summary of the stream along with the total execution time.

Complete Program

```
import time

stream_data = [5, 9, 11, 12, 15]

count = 0
total = 0
start_time = time.time()

for value in stream_data:
    count += 1
    total += value
    running_average = total / count
    print("Received:", value)
    print("Running sum:", total)
    print("Running avg:", running_average)

end_time = time.time()

print("Total elements processed:", count)
print("Final sum:", total)
print("Final average:", total / count)
print("Execution time:", round(end_time - start_time, 6), "seconds")
```

Sample Input

```
stream_data = [5, 9, 11, 12, 15]
```

Expected Output

```
Received: 5
Running sum: 5
Running avg: 5.0
Received: 9
Running sum: 14
Running avg: 7.0
Received: 11
Running sum: 25
Running avg: 8.333333333333334
Received: 12
Running sum: 37
Running avg: 9.25
Received: 15
Running sum: 52
Running avg: 10.4

Total elements processed: 5
Final sum: 52
Final average: 10.4
Execution time: 2e-05 seconds
```

The execution time changes slightly on every run because it depends on the speed of the machine.

Interpretation

- The time module is needed to measure the execution time of the stream.
- Each value arrives sequentially and is processed immediately.
- Each element is processed exactly once.
- The running sum is updated immediately.
- The running average is calculated.
- It shows the total elements processed.
- It shows the final sum.
- It shows the final average.
- It shows the execution time.

Result

The data stream was successfully processed in a single pass. Five elements were processed, giving a final sum of 52 and a final average of 10.4, along with the execution time of the stream.